

Read the passage below, making note of any important points and definitions.

REDUCTIONISM AND EMERGENCE

To illustrate what reductionism means, let's consider an example. Kepler's laws tell us descriptively how the planets move around the sun. Newtonian dynamics tells us, in theory, how any object should move. If we apply Newtonian dynamics to the case of the planets and the sun, we find that Kepler's laws result from the analysis; we are able to explain, in broader terms, why they work. In other words, we have reduced Kepler's laws to a special case of dynamics. Different degrees and levels of reduction can be found in the sciences. In our example, there are two levels of reduction; first, Kepler reduced reams of observational data to a small number of empirical rules, and then Newton reduced these empirical rules to a special case of a general theory. This sort of reduction occurs all the time in the sciences; in a sense, we're really just carrying out a principle aim of science, subsuming the particular to the general. But under what conditions can such a reduction be accomplished? Can any scientific result be reduced in this fashion? If so, isn't all of science then ultimately reducible to some final irreducible theory? These are some of the philosophical questions concerning reductionism. Whether any science can be reduced to some other (more fundamental) science is a hotly debated question.

Let's first consider a classic example of one science being reduced to another. The science of thermodynamics is entirely based on three general laws, which were originally distilled purely from observations and experiments on heat and bulk materials. In other words, thermodynamics says nothing about the microscopic properties of materials (or heat). The science of statistical mechanics, on the other hand, assumes that materials are made of atoms obeying the laws of dynamics. Since we cannot in practice predict the individual motions of the vast numbers of atoms, we use probability theory along with dynamics to make predictions for averaged values. Remarkably, theorists discovered that these averaged values correspond to the variables of thermodynamics. We can now understand thermodynamic results based on a microscopic picture, and so we can say that thermodynamics has been reduced to statistical mechanics. This accomplishment is one of the great victories of reductionism.

But the atoms themselves are made of subatomic particles, which are studied by high-energy physics. Does this mean that thermodynamics can be reduced to high-energy physics? Let's expand this line of thought: Chemistry

studies reactions between atoms and molecules, forming and breaking chemical bonds. These bonds are formed by electrons, subject to the equations of quantum physics. So chemistry is reducible to physics. Biology, however, studies organisms that are made of chemicals. Psychology studies human thinking, which occurs in the brain, a biological organ. Sociology studies societies, which are made of humans. We can then make the claim that sociology is reducible to physics through a kind of hierarchical chain of reductions. There are, however, a number of problems with this absurd claim. An important technical problem is that successful reductions must involve clear and precise correspondences between all terms in the two theories or sciences. If this condition is not satisfied (it rarely is), then each science is an independent thought system with its own terms and structure, not necessarily reducible to anything else. On a practical note, the "reducing" science can't predict all of the results found in the "reduced" science in most cases (not even chemistry and physics, where reductionism seems fairly viable). Favorable synthetic reaction pathways are more often found by working in the lab, not by solving quantum physics equations.

Another argument against extreme reductionism has been advanced by the philosophical movement known as holism. The basic idea here is that a whole may be more than the sum of its parts. A complex system may exhibit emergent properties that can't be predicted by analysis of the component parts of the system. But a reductionist program requires that the complex systems of one science can be understood in terms of the component parts in its "reducing" science. For example, a biological cell must be understandable based solely on the chemical reactions occurring in the cell. If the cell has emergent properties not predicted by these chemical reactions, biology can't be reduced to chemistry. Emergence, as a philosophical doctrine, attracted both interest and controversy (reductionists criticized it as disguised vitalism). Recently, however, the idea of emergent properties has received a boost from within science itself, as the study of complex systems has progressed. In any event, few philosophers (or thoughtful scientists) still subscribe to the radical form of reductionism (although it's still often found in introductory textbooks). The sense in which reductionist programs are still valid (and to what extent) is an active question in the philosophy of science.

Gregory N Derry. *What science is and how it works*. Princeton University Press, 1999.